

8.[Page Replacement Policy]

A system has RAM which can store four pages simultaneously satisfy page requests. Write a Program to calculate percentage pages hit and page fault if the system uses Least recently used page replacement, technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

CODE:

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, faults = 0, hits = 0;
    int flag1, flag2;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag1 = flag2 = 0;

        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                flag1 = flag2 = 1;
                break;
            }
        }

        if(flag1 == 0) {
            for(j = 0; j < framesCount; j++) {
                if(frames[j] == -1) {
                    faults++;
                    frames[j] = pages[i];
                    flag2 = 1;
                    break;
                }
            }
        }

        if(flag2 == 0) {
            int least[10];
            for(j = 0; j < framesCount; j++) {
```

```

GNU nano 8.7
frames[j] = pages[i];
flag2 = 1;
break;
    }
}
if(flag2 == 0) {
    int least[10];
    for(j = 0; j < framesCount; j++) {
        least[j] = -1;
        for(k = i - 1; k >= 0; k--) {
            if(frames[j] == pages[k]) {
                least[j] = k;
                break;
            }
        }
    }
    pos = 0;
    for(j = 1; j < framesCount; j++)
        if(least[j] < least[pos])
            pos = j;
    frames[pos] = pages[i];
    faults++;
}
printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}

```

OUTPUT:

```

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ cd /c/Users/Rachna/Desktop
-bash: cd: /c/Users/Rachna/Desktop

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ nano lru.c

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ gcc lru.c -o lru

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ ./lru
Enter number of pages: 7
Enter page reference string:
1 2 3 2 4 1 5
Frame status: 1 -1 -1
Frame status: 1 2 -1
Frame status: 1 2 3
Frame status: 1 2 3
Frame status: 4 2 3
Frame status: 4 2 1
Frame status: 4 5 1

Total Page Hits = 1
Total Page Faults = 6
Hit Percentage = 14.29%
Fault Percentage = 85.71%

rachn@LAPTOP-6NDD8TJ3 MSYS ~
$ |

```